

Work these problems to prepare for Quiz 5 (Sections 14.6–14.7). Show your work. Solutions are on the back.

1. Let $f(x, y) = x^2y - 3y^2$.
 - (a) Find the directional derivative of f at the point $(1, 2)$ in the direction toward the point $(4, 6)$.
 - (b) Find the maximum rate of change of f at $(1, 2)$ and the unit direction in which it occurs.

2. Find the equation of the tangent plane to the surface $x^2 + y^2 - z^2 = 4$ at the point $(2, 1, -1)$.
Hint: Write $F(x, y, z) = x^2 + y^2 - z^2$ and use ∇F .

3. Find all critical points of $f(x, y) = 3x^2 + x^2y + y^3 + 6y^2$ and classify each as a local maximum, local minimum, or saddle point.

4. Find the absolute maximum and absolute minimum values of $f(x, y) = x + y - xy$ on the triangular region D with vertices $(0, 0)$, $(6, 0)$, and $(0, 6)$.

5. Find the point on the plane $3x + 2y + z = 14$ that is closest to the origin.

Hint: Minimize $D^2 = x^2 + y^2 + z^2$ after eliminating z .